Andrew J. Peng

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EDUCATION

Bachelor of Science in Computer Engineering — Expected Fall 2028
University of Maryland — College Park, MD

- **GPA**: 3.96 | A. James Clark School of Engineering Dean's List: F24, S25, F25, S26

EXPERIENCE

Game Developer — Dec. 2024 – Present
Contractor — Belmont, CA

- Contracted for Tiny Lab Studio; architected the **server-authoritative combat framework** for **Push Party**, a live multiplayer Roblox game with **70,000+ visits**, **25,000+ unique players**, and up to **100 concurrent players**.
- Built exploit prevention with **server-side validation** of combat actions, **rate limiting** on remote events, and **server-authoritative movement checks** that detect unauthorized flight and correct player positions.
- Utilized **Luau** to design and deliver **30+ modular game systems**, including data management, UI/UX, gameplay mechanics, and physics simulations for client commissions on **Roblox**.

Machine Learning Research Assistant — May 2026 – Aug. 2026
Shanghai Jiao Tong University — Shanghai, China

- Researched **few-shot object detection** on a **4,481-image, 25-class** satellite dataset of ships, military aircraft, and launch vehicles under **Prof. Hongtian Chen**, working with graduate researchers.
- Benchmarked **RF-DETR Small** in **PyTorch** with **COCO-style** evaluation, **worst-image ranking**, and **SAHI/tiled inference** to find the most reliable strategy for **small, few-shot** classes.
- Debugged model accuracy errors through per-class and confusion-matrix analysis, tracing them to **dataset labeling inconsistencies**; used **DarkLabel** to review and correct mislabeled annotations.

Vehicle Dynamics & Ergonomics Engineer — Aug. 2024 – May 2026
Terps Racing — College Park, MD

- Designed **ABS housings** using **SolidWorks** and integrated **brake temperature sensors** and **push rod strain gauges** for the 2026 FSAE EV; validated sensor mounts with FEA against vibrational racing loads.
- Modeled a **clutch actuation** assembly in **SolidWorks** for the 2025 FSAE IC vehicle, validating the carbon lever and aluminum mount with FEA under **150 N peak input**; manufactured parts by **water jet** and **3D printing**.

PROJECTS

LiteLabel — Sept. 2026 – Oct. 2026

- Built a cross-platform **bounding-box image annotation tool** in **Python** and **PySide6** that reads and writes **YOLO**, **RF-DETR**, **Pascal VOC**, and custom label formats, each defined by a field-based **YAML** schema.
- Wrote **100+ pytest tests** and a **GitHub Actions** release pipeline that runs them before building standalone **Windows** and **macOS** apps; used **atomic writes** so a failed save never corrupts labels.

ProfileLite — Dec. 2025 – Jan. 2026

- Reimplemented the session-locking design of **ProfileService** and **ProfileStore** from scratch as one **strict-typed** Luau library that keeps player data consistent across **concurrent game servers**.
- Wrote a **force-load handoff** over **MessagingService**, **atomic UpdateAsync** writes with **backoff retries**, a cross-server message queue, version rollback, and an offline **mock store**; published on the Roblox Creator Store.

SKILLS

Languages: Python, Luau, Java, MATLAB, HTML/CSS
Tools: Git, GitHub Actions, pytest, Apache Subversion, Roblox Studio, DarkLabel, Arduino, Raspberry Pi, SolidWorks
Frameworks: PyTorch, PySide6 (Qt), OpenCV, NumPy, Pandas, Matplotlib, RF-DETR, YOLOv8